UCLA Undergraduate
RESEARCH WEEK

Tracking the Spread: Spatiotemporal Shifts in Fentanyl Overdose Deaths Before and After COVID-19 in Los Angeles County

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County of Los Angeles

The Fentanyl Crisis & Why Now?

- Opioid crisis has intensified; fentanyl leading overdose deaths and contamination
- COVID-19 presumably exacerbated overdose risk: isolation, instability, reduced access to care
- Current public health interventions are **reactive**

The Question

Can we predict overdose surges at the ZIP-code level in LA County?

Approach: Use data science to detect patterns and build predictive tools that enable <u>local</u>, <u>proactive</u> intervention among public health workers

The Data

Coroner records of overdose deaths in Los Angeles County from 2012–2024.

Key Features

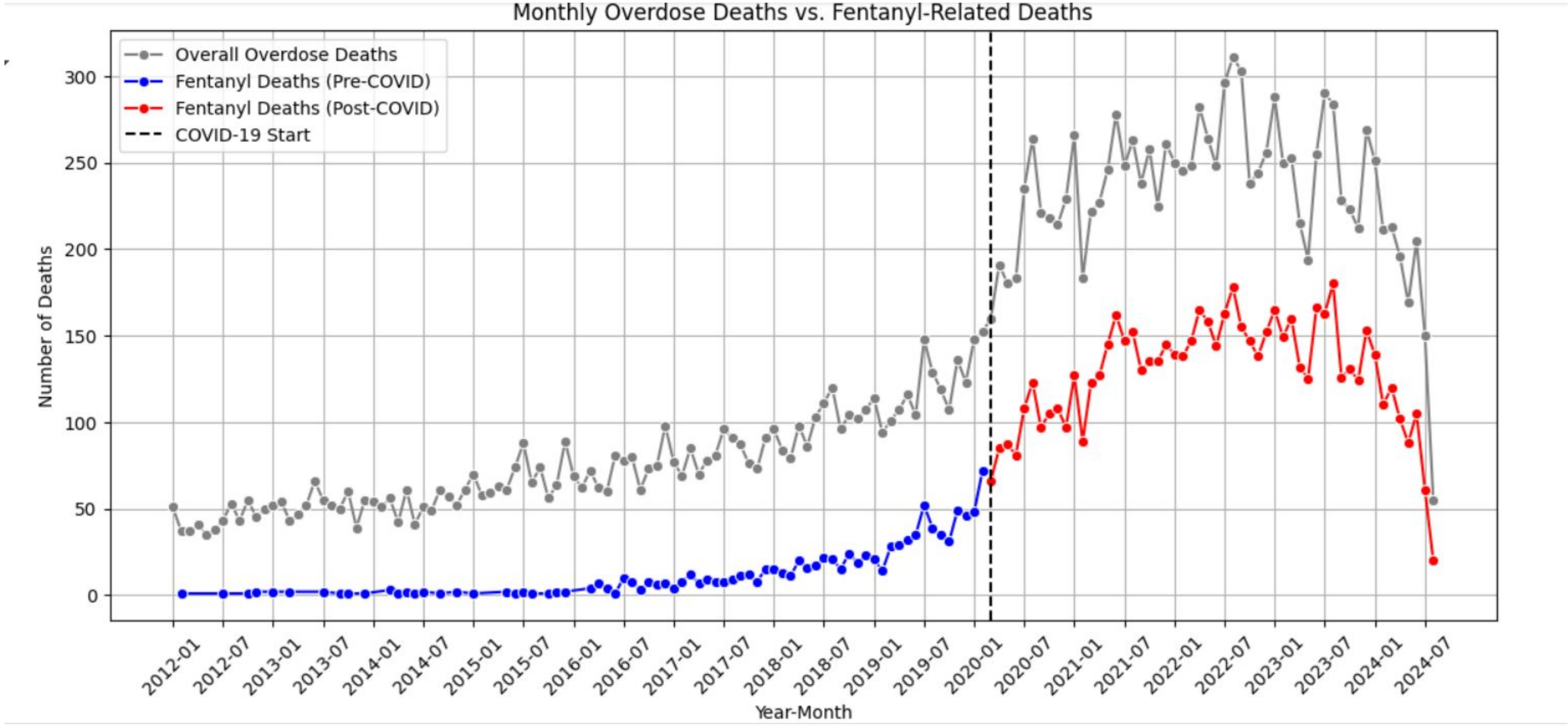
Date of Death

Location Information

Demographics

Cause of Death (Fentanyl)

Evidence of Post-COVID Surge



Fentanyl overdose increase demonstrated against all drug overdoses.

Evidence of Post-COVID Surge

- Marked increase in fentanyl deaths after COVID onset (March 2020)
- Not necessarily causal; correlation
- Could this surge have been **anticipated** and **mitigated** with better surveillance tools?

Mapping the Spread

Mapping the Spread

- Post-COVID surge increased deaths where incidence was already high
- New ZIP codes with little to no prior incidence seeing overdoses
- Heatmaps offer powerful visual insight — but can we move from visualization to **prediction**?

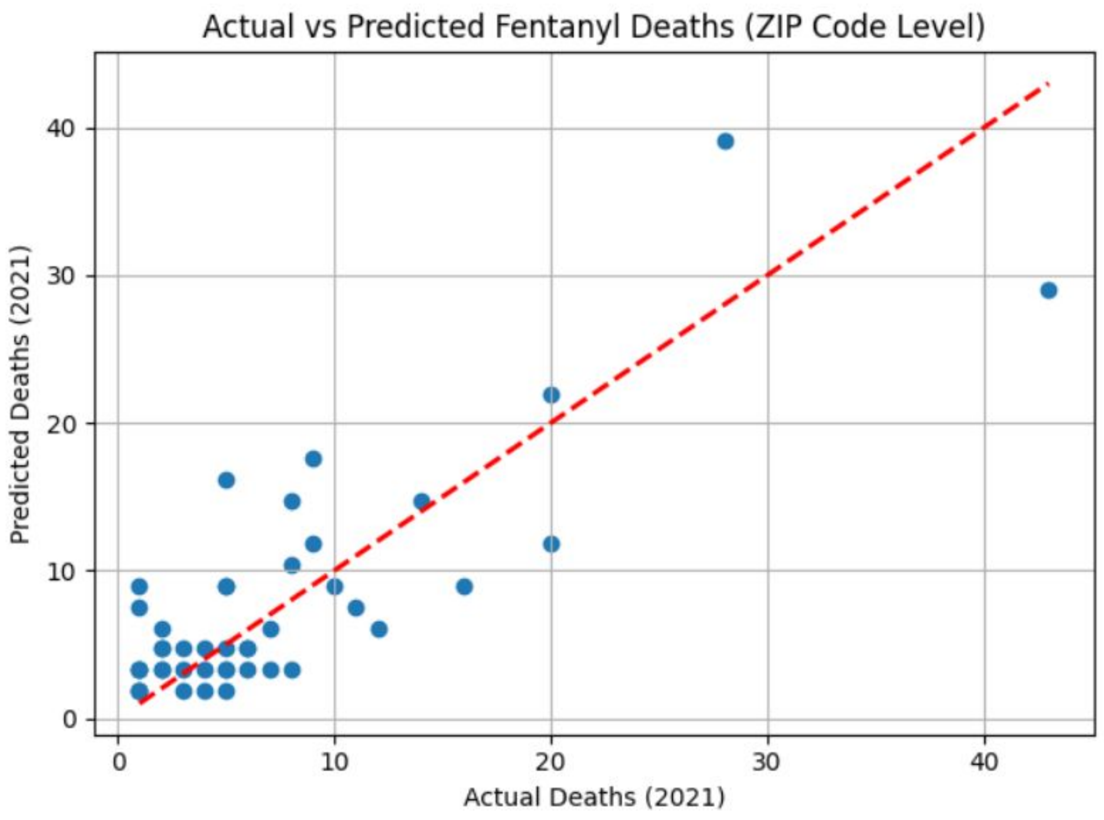
Analysis Overview

- Linear Regression
- XGBoost Classification
- Interactive ZIP-code Dashboard

Does the Data Allow Prediction?

- **Linear Regression**
- **Input:** 2020 ZIP-level fentanyl data
- **Output:** 2021 ZIP-level data
- **Performance:** $R^2 = 0.73$
- **Interpretation:** 73% of variation in 2021 deaths explained by 2020 data

Limitation: Don't capture sudden shifts or nonlinear effects



Can We Predict Monthly Surges?

- **XGBoost Classifier** – uses regression to support classification
- **Inputs:** Prior 6 months’ overdose counts (lag features), ZIP-level demographics, cyclic time features
- **Output:** Binary classification — Surge (increase) vs. No Surge
- Dataset balanced to address class imbalance (fewer surges)
- **Decision threshold:** 0.4, tuned to maximize **recall** — favoring early warnings

Can We Predict Monthly Surges?

	Predicted: No Surge (0)	Predicted: Surge (1)
Actual: No Surge (0)	178	11
Actual: Surge (1)	17	148

Confusion Matrix

Metric	Class 0 (no surge)	Class 1 (surge)
Precision	91%	93%
Recall	94%	90%
F1-score	93%	91%

Classification Report

- Confusion matrix → true surges correctly identified
- False positives occur in noisier ZIPs
- **Limitation:** Model performance may degrade if future overdose patterns shift dramatically due to new external factors

From Model to Tool: Risk Dashboard

Inputs: Classifier model, Standard scalar to normalize features, classifier data

Predicting a fentanyl surge for the month **after the latest recorded month** in the dataset for each ZIP code

Dashboard’s Users

- LA County epidemiologists and surveillance staff
- Harm reduction program managers and first responders
- Public health nurses, mobile clinics, and community health workers
- Decisions around Narcan deployment, outreach prioritization, resource planning

Key Takeaways

- Post-COVID overdose surge is visible, mappable, and **potentially predictable**
- Time-lagged data + machine learning can give early signals
- Visual and interactive tools can support real-time public health decision-making

Future Directions

- Expand dashboard with alert system, local facilities (Narcan pop-ups)
- Pilot collaboration with LA County health agencies or clinics
- Generalize models to other high-risk substances and counties
- Incorporate broader social determinants (e.g., unemployment)

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My Code

https://github.com/hritikac25/fentanyl_overdose_covid